

5.18 Solution: (a) Let us assume that the z -axis is perpendicular to the surface of the slab. For the ring O_1 in the vacuum parallel to the slab located at $z = d$, according to Problem 5.17, it will induce an image current ring O_2 in the slab at $z = -d$, with an image current of

$$I' = \frac{\mu - 1}{\mu + 1} I,$$

and in the same direction as O_1 . Therefore, according to Section 5.5, the vector potential generated by O_2 on O_1 is a constant, and is in the same direction as I . With $r = \sqrt{a^2 + 4d^2}$ and $r \sin \theta = a$, the vector potential is

$$A_\phi = \frac{\mu_0 I'}{2\pi} \frac{a}{\sqrt{a^2 + d^2}} \left[\frac{(2 - k^2)K(k) - 2E(k)}{k^2} \right] = \frac{\mu_0 I'}{2\pi} \left[\left(\frac{2}{k} - k \right) K(k) - \frac{2}{k} E(k) \right],$$

with

$$k = \frac{a}{\sqrt{a^2 + d^2}}.$$

The interaction energy between O_1 and O_2 is then given by

$$W = \oint \mathbf{J} \cdot \mathbf{A} dl = \mu_0 I I' a \left[\left(\frac{2}{k} - k \right) K(k) - \frac{2}{k} E(k) \right].$$

The force acted on O_1 can be calculated as the force between O_1 and O_2 , which is, by Eq. (5.150),

$$F = \frac{\partial W}{\partial (2d)} = \frac{\mu_0 I I' a}{2} \left[\left(-\frac{2}{k^2} - 1 \right) K(k) + \left(\frac{2}{k} - k \right) \frac{dK(k)}{dk} + \frac{2}{k^2} E(k) - \frac{2}{k} \frac{dE(k)}{dk} \right] \frac{\partial k}{\partial d}.$$

Since

$$\frac{dK(k)}{dk} = \frac{E(k)}{k(1 - k^2)} - \frac{K(k)}{k}, \quad \frac{dE(k)}{dk} = \frac{E(k) - K(k)}{k}, \quad \text{and} \quad \frac{\partial k}{\partial d} = -\frac{ad}{(a^2 + d^2)^{3/2}},$$

the force now becomes

$$F = -\frac{\mu_0 I I' a}{2} \left[-\frac{2}{k^2} K(k) + \frac{1}{1 - k^2} \left(\frac{2}{k^2} - 1 \right) E(k) \right] \frac{ad}{(a^2 + d^2)^{3/2}}.$$

(b) With O_1 perpendicular to the surface of the slab, the induced image current ring is as shown in Fig. (). Move the origin of the coordinate system to the center of O_2 , then the magnetic field generated by O_2 in the xz plane is given by, from Section 5.5, with θ measured from the y -axis and $\cos \theta = 0$ for the xz plane,

$$B_r = \frac{\mu_0 I'}{2r} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{a^{2n+1}}{r^{2n+2}} P_{2n+1}(\cos \theta = 0) = 0,$$

as $P_{2n+1}(0) \equiv 0$, and

$$B_\theta = -\frac{\mu_0 I'}{4} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n (n+1)!} \frac{a^{2n+2}}{r^{2n+3}} P_{2n+1}^1(\cos \theta = 0),$$

where

$$r^2 = a^2 + 4d^2 + 4ad \cos \beta.$$

Since

$$P_{2n+1}^1(0) = -\frac{(-1)^n (2n+1)!}{2^{2n} (n!)^2},$$

we can see that

$$B_\theta = \frac{\mu_0 I'}{4} \sum_{n=0}^{\infty} \frac{(2n+1)!(2n+1)!!}{2^{3n}(n+1)!(n!)^2} \frac{a^{2n+2}}{r^{2n+3}}.$$

The force can now be directly calculated with

$$\mathbf{F} = \oint d\mathbf{I} \times \mathbf{B}.$$

Here,

$$d\mathbf{I} = Ia(-\hat{x} \cos \beta + \hat{z} \sin \beta) d\beta, \quad \mathbf{B} = \hat{\theta} B_\theta = -B_\theta \hat{y},$$

we will have

$$\mathbf{F} = Ia \int_0^{2\pi} (\hat{z} \cos \beta + \hat{x} \sin \beta) B_\theta d\beta.$$

There is no x -component, as the integrand is odd. Finally, the force is in the z -direction, with

$$F = \frac{\mu_0 II'}{4} \sum_{n=0}^{\infty} \frac{(2n+1)!(2n+1)!!}{2^{3n}(n+1)!(n!)^2} a^{2n+3} \int_0^{2\pi} \frac{\cos \beta}{(a^2 + 4d^2 + 4ad \cos \beta)^{(2n+3)/2}} d\beta.$$

(c) For $d \gg a$, we have $k \sim 0$, we can use the MacLaurin series of $K(k)$ and $E(k)$,

$$K(k) = \frac{\pi}{2} \left(1 + \frac{k^2}{4} + \frac{9}{64} k^4 \right), \quad E(k) = \frac{\pi}{2} \left(1 - \frac{k^2}{4} - \frac{3}{64} k^4 \right),$$

to get the force in the parallel case. (Note that, Gradshteyn and Ryzhik, 8th ed., 8.114.1, pg. 870, is wrong, see DLMF or Wiki.)

Now, the force becomes

$$\begin{aligned} F &= -\frac{\mu_0 II' a}{2} \left[-\frac{2}{k^2} \cdot \frac{\pi}{2} \left(1 + \frac{k^2}{4} + \frac{9}{64} k^4 \right) + \left(\frac{2}{k^2} + \frac{1}{1-k^2} \right) \cdot \frac{\pi}{2} \left(1 - \frac{k^2}{4} - \frac{3}{64} k^4 \right) \right] \frac{ad}{(a^2 + d^2)^{3/2}} \\ &= -\frac{\mu_0 \pi II' a}{2} \frac{3k^2}{8} \frac{ad}{(a^2 + d^2)^{3/2}}. \end{aligned}$$

Then, using $k \approx a/d$, we will finally have

$$F = -\frac{\mu_0}{2\pi} \left(\pi a^2 I \right) \left(\pi a^2 I' \right) \frac{3}{(2d)^4}.$$

For a magnetic dipole $\mathbf{m}_1$, the magnetic field generated by it is

$$\mathbf{B}_1 = \frac{\mu_0}{4\pi} \frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{m}_1) - \mathbf{m}_1}{r^3}.$$

The interaction energy between this magnetic dipole moment and another $\mathbf{m}_2$ is therefore

$$W = \mathbf{m}_2 \cdot \mathbf{B}_1 = \frac{\mu_0}{4\pi} \frac{3(\mathbf{n} \cdot \mathbf{m}_1)(\mathbf{n} \cdot \mathbf{m}_2) - \mathbf{m}_1 \cdot \mathbf{m}_2}{r^3},$$

and the force between them is

$$F = \frac{\partial W}{\partial r} = -\frac{\mu_0}{4\pi} \frac{3[3(\mathbf{n} \cdot \mathbf{m}_1)(\mathbf{n} \cdot \mathbf{m}_2) - \mathbf{m}_1 \cdot \mathbf{m}_2]}{r^4}.$$

In the parallel case, $\mathbf{m}_1 = \pi a^2 I \hat{z}$, $\mathbf{m}_2 = \pi a^2 I' \hat{z}$, $\mathbf{n} = \hat{z}$, and $r = 2d$, and the force is

$$F = -\frac{\mu_0}{4\pi} \frac{6(\pi a^2 I)(\pi a^2 I')}{(2d)^4},$$

which agrees with the limiting form of part (a).

We can perform the similar calculation for the perpendicular case. To this end, we only need to retain the $n = 0$ term, and the force is

$$\begin{aligned} F &= \frac{\mu_0 I I'}{4} \left(\frac{a}{2d}\right)^3 \int_0^{2\pi} \cos \beta \left(1 + \frac{a}{d} \cos \beta + \frac{a^2}{4d^2}\right)^{-3/2} d\beta \\ &= \frac{\mu_0 I I'}{4} \left(\frac{a}{2d}\right)^3 \int_0^{2\pi} \left(\cos \beta - \frac{3a}{2d} \cos^2 \beta\right) d\beta \\ &= -\frac{\mu_0 \pi I I'}{4} \left(\frac{a}{2d}\right)^3 \frac{3a}{2d}. \end{aligned}$$

Using dipole approximation, we have $\mathbf{m}_1 = -\pi a^2 I \hat{y}$, $\mathbf{m}_2 = \pi a^2 I' \hat{y}$, $\mathbf{n} = \hat{z}$, and $r = 2d$, and the force is

$$F = -\frac{\mu_0}{4\pi} \frac{3(\pi a^2 I)(\pi a^2 I')}{(2d)^4},$$

which also agrees with the limiting form of part (b).